Probing DNA Damage Repair Mechanisms

A thesis submitted to the University of Dublin, Trinity College for the degree of B.A. (Mod) Chemistry with Biosciences



By Bláthnáit Alice Logan

Based on research carried out under the supervision of Prof. Joanna McGouran

Michaelmas Term 2023

Student Number: 20332855

I have read and understand the plagiarism provisions in the General Regulations of the University Calendar which are found at http://www.tcd.ie/calender

I have also read and completed the Online Tutorial on avoiding plagiarism 'Ready Steady Write' which can be found at http://tcd.ie-libguides.com/plagarism/ready-steady-write

This work is entirely my own and all information and work used has been acknowledged as such.

Signed by:

Bláthnáit Logan

Acknowledgements

I would like to thank Prof. Joanna McGouran and Connor O'Leary for their support, supervision and guidance throughout the course of this project. I would also like to thank the McGouran Group- Susie, André, Ellen, Denise, Valerio, Sean and Tomasz for their help, advice and laughs throughout my time in the lab. A huge thank you also to Dr John O' Brien and Dr Manuel Ruther for their invaluable knowledge regarding NMR. I would also like to thank my family Siobhán, Paul and Naoise for being a constant support and supporting me in all my endeavours. Sending all my love and thanks to my friends for being the best cheerleaders throughout my degree and especially to Kevin for the constant encouragement and support. And finally, to my aunt Ann Marie for always giving the best advice even if I don't particularly want to hear it.

Abbreviations

A	Adenine
ABP	Activity-based probe
ABPP	Activity-based protein profiling
Ac	acetyl
Ac_2O	Acetic anhydride
app	Apparent
approx.	Approximately
B	Nucleobase
BAIB	(diacetoxyiodo)benzene
Bn	Benzyl
Br	Broad
Bs	Broad Singlet
Bz	Benzoyl
C	Cytosine
COSY	Correlated spectroscopy
Cyt	cytidine
d	doublet
dd	doublet of doublets
DCM	Dichloromethane
DMF	N, N'- Dimethylformamide
DMSO	Dimethyl Sulfoxide
DMP	Dess-Martin Periodinane
DMTr	4,4'-Dimethoxytrityl
DNA	Deoxyribonucleic Acid
ESI	Electrospray ionisation
Et_2O	Diethyl Ether
EtOAc	Ethyl acetate
EtOH	Ethanol
FA	Fanconi Anemia
G	Guanine
Hex	Hexane

HF	Hydrogen Fluoride
HMBC	Heteronuclear multiple bond correlation
HRMS	High resolution mass spectroscopy
HSQC	Heteronuclear single quantum correlation
Hz	Hertz
ICL	Interstrand Crosslink
IR	Infrared
J	Coupling constant
m	multiplet
MeOH	Methanol
m.p.	Melting Point
m/z	Mass to charge ratio
NMR	Nuclear magnetic resonance
Ph	Phenyl
ppm	Parts per million
Pso2	Sensitivity to psoralen 2
R_f	Retention factor
RNA	Ribonucleic Acid
Rt	Room temperature
s	singlet
SNM1A	Sensitivity to Nitrogen Mustard 1 A
t	triplet
T	Thymine
TBAF	Tetra-n-butylammonium fluoride
TBDMS	Tert-butyldimethylsilyl
TOCSY	Total correlation spectroscopy
THF	Tetrahydrofuran
TLC	Thin layer chromatography
U	Uracil
UV	Ultraviolet
ZBG	Zinc Binding Group
Zn	Zinc

Abstract

Human SNM1A is a 5’ to 3’ exonuclease that plays a significant role in various DNA damage repair processes. It digests sections of damaged DNA along a 5’-3’ directionality and enables subsequent repair. SNM1A and other members of the SNM1 gene family are poorly characterised, hence the majority of its roles have not been fully elucidated. It appears to have a high affinity for digesting interstrand cross link lesions. Malfunctioning SNM1A can cause defective DNA damage repair, leading to an increased incidence of cancer and genetic conditions such as Cockayne Syndrome and Fanconi Anemia. Therefore, investigating SNM1A activity would be of great interest in aiding the development of therapeutics for chemotherapy resistant cancers.

In order to further explore the activity of this enzyme, a multistep organic synthesis was followed to create modified mononucleosides. These mononucleosides are intermediates in the construction of a novel mononucleotide which will be used to build an activity-based probe. The mononucleosides shall be joined to an oligonucleotide strand by solid phase synthesis for testing against SNM1A. The intermediates are attached through a 2’-5’ linkage rather than the standard 3’-5’ linkage in order to provide resistance to hydrolysis.

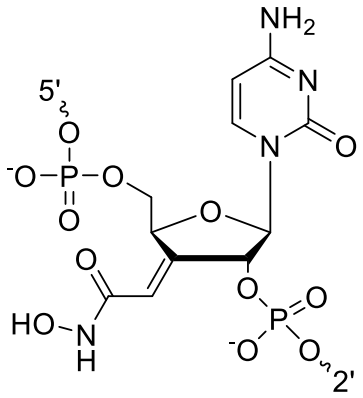


Figure 1: Phosphodiester bonds attached through the 2’ and 5’ linkage

NOTE: The labelling of the nucleoside in this thesis is denoted as follows

Within this report nucleosides adhere to standard nucleoside naming convention as displayed below. Atoms that belong to the nucleobase are labelled 1-6 whilst the atoms on the deoxyribose sugar are labelled from 1'-5'.

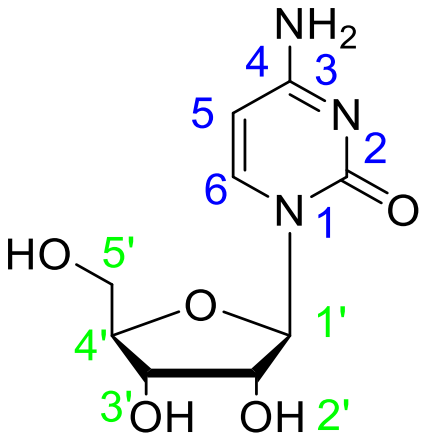


Table of Contents

Acknowledgements i

Abbreviations ii

Abstract iv

Explanatory Note v

1. Introduction

1.1 DNA damage and repair 1

1.2 Interstrand Crosslinks and the Fanconi Anemia Pathway 2

1.3 SNM1A and its role in DNA damage repair 3

1.4 Reported inhibitors of SNM1A and zinc binding group incorporation 6

1.5 Oligonucleotide synthesis and modification 8

1.6 Activity based Probes 9

1.7 Aims and Objectives 10

2. Results & Discussion

2.1 Synthesis Strategy of Target Compounds and Protecting Group Implementation 12

2.2. Benzoylation of the Exocyclic Amine 14

2.2.1 2'- and 5'- Protection and Oxidation 16

2.2.2 Protection of the Exocyclic Amine 19

2.3 Revised Synthetic Strategy 20

3. Conclusions and Future Work 22

4. Experimental Methods 24

5. References 33

6. Appendix 37

1. Introduction

1.1 DNA Damage and Repair

Deoxyribonucleic acid (DNA) exists in living cells and encodes the genetic information, required for the function and development of an organism.[1]It is a polymer composed of two linked strands that wind around one another to produce a double helix. Each strand is composed of nucleotides – units made from a sugar molecule, a phosphate group and a nitrogenous base (A, T, C, G). The sugar phosphate backbone provides support and protects structural integrity[2.] When the function of DNA is impaired, it prevents transcription and replication from taking place. The misfunction of these fundamental operations leads to negative consequences for the genome as a whole- resulting in phenomena such as genomic instability[3] and apoptosis.[3] DNA is constantly susceptible to damage by both exogenous and endogenous factors. Exogenous factors or environmental factors include ionizing radiation (IR)[3], UV light[3] and various chemical substances. Endogenous DNA damage can be caused by hydrolysis[3], alkylation[3], oxidation[3] and mismatching of DNA bases during replication.[3] Whilst the cell has several innate repair mechanisms that target DNA mutations and replication errors, DNA damage can still evade these processes, with negative consequences for the organism. An especially lethal form of DNA damage occurs when two opposing strands of the duplex become covalently bonded together- creating an interstrand crosslink.

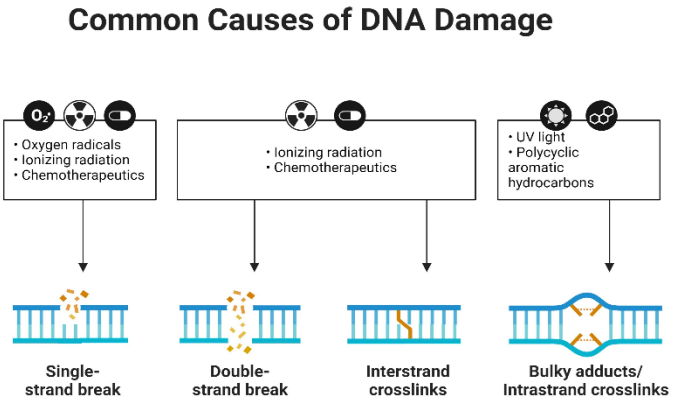


Figure 1: Common Causes of DNA Damage[4]

1.2 Interstrand Crosslinks and the Fanconi Anaemia Pathway

ICLs (interstrand crosslinks) are highly cytotoxic covalent lesions that connect two nucleotide residues from the same DNA strand. [5] ICLs are particularly toxic as the complementary strands of the DNA duplex are covalently bonded together.[5] These covalent bonds prevent the unwinding of DNA, thus preventing DNA strand separation, an important function required for activities such as transcription and replication. The chemical structure of the resulting ICL depends on the crosslinking agent implicated, and these different structures will lead to different cellular responses.[9] ICLs can be formed exogenously by naturally occurring compounds such as psoralen and mitomycin C and by synthetic bifunctional alkylators such as cisplatin.[6,7] Similarly, they can be created endogenously by lipid peroxidation[7] and reactive aldehydes.[8] Due to its extreme genotoxic and cytotoxic nature, one unrepaired DNA ICL could yield lethality in monocellular organisms, whereas 20 to 40 unrepaired ICLs are fatal to mammalian cells.[24,25] Mitomycin C and cisplatin, commonly used chemotherapeutic agents , utilize the potent cytotoxicity of ICLs, thus creating crosslinks in tumour cells in order to disrupt the process of DNA synthesis in the tumour. [9] Certain cancers and genetic disorders however may acquire resistance to crosslinking agents as a result of overexpression of the enzymes involved in these ICL repair pathways.[13]

Fanconi Anemia (FA) is an autosomal recessive genetic disorder brought upon by faulty DNA repair of ICLs.[11,14] A typical feature of cells derived from FA patients is their hypersensitivity to ICL anti-tumour agents[14] such as mitomycin C and cisplatin, as they possess a defective DNA repair pathway responsible for repairing these lesions. The genomic instability of FA patients is closely correlated with cancer development. Patients have reduced survival rates, susceptibility to tumour formation and elevated chromosomal abnormalities. In the case of FA, it has been deduced that mutations in 14 genes[16] associated with the resolution of ICL's during replication allow for manifestation of this disorder. As a result, ICL repair mechanisms play a

fundamental role in maintaining genomic integrity and enabling cell survival. In eukaryotic cells, ICL lesions are recognised and repaired during the S phase which occurs between the G1 and G2 phases of the cell cycle.[22] Replication forks collide with the ICL, which allow for subsequent repair of the lesion.[22] In a study by Mason and Sekiguchi, [15] they found that SNM1B, an ortholog of the Snm1/Pso2 gene, plays a critical role in ICL repair and when this gene expression is maimed, the ICL repair function is impaired. Thus, establishing a connection between impaired DNA damage repair enzymes and ICL repair.

The human enzyme SNM1A which is another mammalian ortholog of the Snm1/Pso2 gene has been shown to initiate a repair response against ICLs.[15] It does this by making the starting incision in the damaged DNA, cleaving the crosslinks and creating a starting point for proteins to mediate the repair process. Cells depleted in SNM1A show increased sensitivity to certain chemotherapeutics such as cisplatin.[17,18] As SNM1A has been demonstrated as a strong candidate for potential targeting of chemotherapy resistant cancers, developing SNM1A inhibitors leads to the possibility of more effective therapeutic outcomes and improved patient prognosis.

1.3 SNM1A and its role in DNA damage repair

Human SNM1A, also known as SNM1 (sensitivity to nitrogen mustard 1), is a 5'- to -3' exonuclease that aids with replication-dependent and transcription-coupled interstrand crosslink repair.[17,35] SNM1A has captured scientific interest due to its specificity for repairing damage caused by ICLs. Whilst little is known about the enzyme, it has been confirmed that it possesses exonuclease activity, allowing it to recognise and cleave foreign DNA such as ICLs on the defined recognition sites.[18]SNM1A is characterised by metallo-β-lactamase (MBL)[17] and β-CASP domains which act together to generate nuclease activity.[17,18]



. **Figure 2:** Crystal structure of SNM1A [19]

The crystal structure of SNM1A has been elucidated and bears similarity with other mammalian orthologs of the SNM1 gene family - SNM1B and SNM1C. SNM1A has a positively charged surface due to the presence of side chains composed of lysine and arginine residues located on both the MBL and β-CASP domains.[13]This positively charged face undergoes electrostatic interactions with the negatively charged phosphodiester DNA backbone, this repulsion allows for the nucleotides to fold and partake in biomolecular interactions such as binding to DNA proteins.[45]

The SNM1 family possesses an extended active site where the MBL fold domain allows for zinc ions present in the active site to bind to the enzyme, promoting enzyme activity. The MBL fold displays a preference for single stranded DNA *in vitro* and does not possess the capacity to digest single- or double-stranded RNA.[20] A zinc binding pocket is present in the centre of the structure that defines the selectivity for the terminal 5' phosphate.[22] The wide, positively charged DNA binding groove is essential for enzyme activity as it enables the enzyme to digest former sites of DNA degradation.[20] A phosphate binding pocket present facilitates phosphodiester hydrolysis.[35,37]

SNM1B differs from SNM1A as it has a di-zinc catalytic centre, whilst SNM1A only possesses a singular zinc atom in its centre,[17] as can be seen in **Fig. 2.** Whilst it has been postulated that

SNM1A contains a second zinc ion in its active site or another transition metal cation that is more loosely bound,[13] no crystallographic evidence has been found to support this claim. If this were the case, the two zinc ions would activate a water molecule which subsequently would carry out a nucleophilic attack on a phosphodiester in the DNA backbone and stabilise the negative charge on the phosphodiester as it underwent hydrolysis. Following hydrolysis, the oligonucleotide would be reduced in size by one residue, which allows for SNM1A to continue digestion of the DNA strand. As this hypothesis is yet to be proven, it is taken that the singular zinc ion in the active site is held in place by three histidine residues. The residues implicated in the coordination of the zinc ions in SNM1B lie in the same positions as SNM1A, except for a lone aspartate residue. SNM1A has complete dependence on the aspartate residue (D736) for nuclease activity that should coordinate the missing zinc ion.[20,21] This residue's importance has been investigated by Allerston *et al.*[17,20], who proved that when the aspartate is replaced with the amino acid alanine, the enzyme shows a large decrease in catalytic activity.

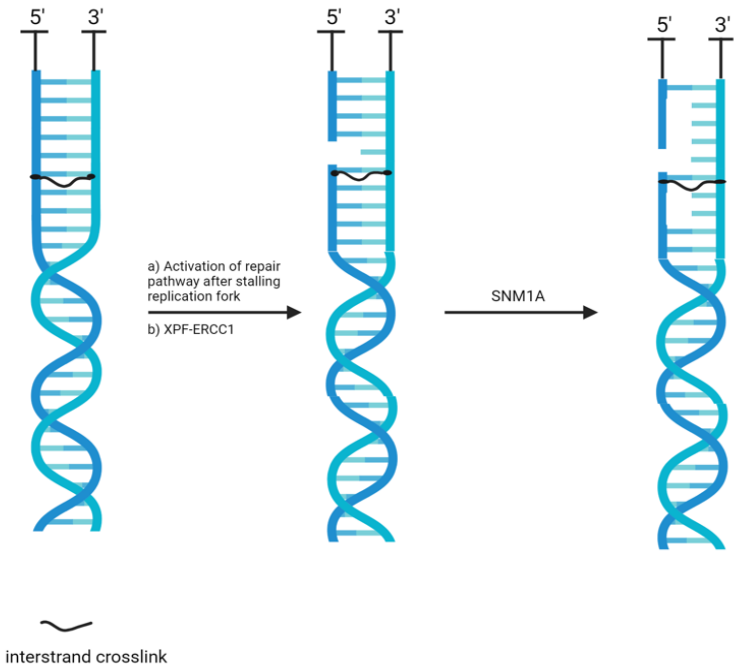


Figure 3: Key steps in SNM1A DNA repair showing SNM1A digesting past the ICL.[43]

As mentioned previously, SNM1A has the propensity to digest past ICLs making it an attractive target for chemotherapy resistant cancers of this type. It is able to carry out this process due to the epistatic relationship it has with XPF-ERCC1.[21] XPF-ERCC1 is a structure specific endonuclease that makes the initial 5' strand incision before the ICL.[38] This incision provides a port of entry for SNM1A, which then hydrolyses one strand of DNA past the point of the ICL, unhooking the lesion and leaving a lone nucleotide tethered to the complementary DNA strand **(Fig. 3).** [35,36] DNA repair can be concluded by a variety of different processes including trans-lesion synthesis,[37] nucleotide excision repair (NER)[38] and homologous recombination.[48] Many chemotherapeutic agents function by inducing cytotoxic DNA damage, however, over time tumour cells become resistant, in part by upregulation of DNA repair pathway.

1.4 Reported inhibitors of SNM1A and zinc binding group incorporation

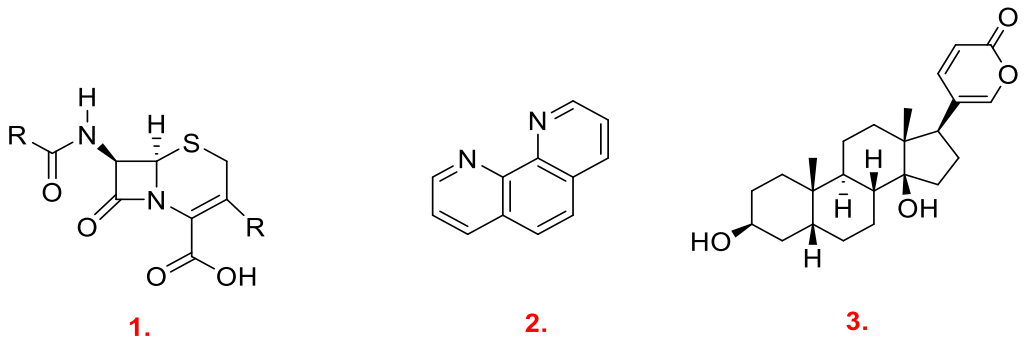


Figure 4: Reported inhibitors of SNM1A 1. Cephalosporin, 2. *o*-phenanthroline 3. Bufalin

SNM1A inhibitors can be useful as adjunctive chemotherapeutic agents for tumour cells that have lost DNA repair functionality. Various inhibitors for SNM1A have been identified that inhibit nuclease activity *in vitro* at low micromolar efficiency.[27] Three of the most prominent inhibitors are cephalosporins, the zinc binder σ-phenanthroline and the steroid toxin bufalin **(Fig. 4)**. In a study performed by Sengerová *et al.*,[20] it was shown that SNM1A activity is weakened once the concentration of *o*-phenanthroline is increased. As SNM1A is part of the metallo-β-lactamase superfamily of enzymes, it is inhibited by cephalosporins which are a class

of β-lactam antibiotics. The most potent inhibitors of this class are the third generation cephalosporins- cefotaxime[27] and ceftriaxone[27], they possess IC^{50} values in the 4-7 μM range and operate as competitive reversible inhibitors.[27]Bufalin was reported to have *in vitro* anti-tumour effects as demonstrated by Wu *et al.*,[28] who showed that bufalin induces apoptosis in human lung cancer cells. Bufalin alters expression levels of numerous genes involved in DNA repair and cell cycle regulation, including SNM1A.

Zinc ions are the driving force for the catalytic activity of various enzymes. As reported by McCall and Huang, [46] zinc is fundamental for the function of over 300 enzymes and is found in all six classes of enzyme- oxidoreductases, transferases, hydrolases, lyases, isomerases, and ligases. To create a modified nucleoside inhibitor for SNM1A, a zinc binding group (ZBG) can be incorporated into the nucleoside to mimic the original phosphate group whilst creating a steady interaction with Zn^{2+} in the binding site of SNM1A. Zn^{2+} is suspected to fix inhibitors within the active site whilst hydrolysing oligonucleotides in the presence of phosphodiesterase's **(Fig. 5)**.[33] Although the actual mechanism is unknown, it is suspected to be similar to MBL enzymes which rely on a bridging hydroxide to attach the two Zn^{2+} ions together. [20]

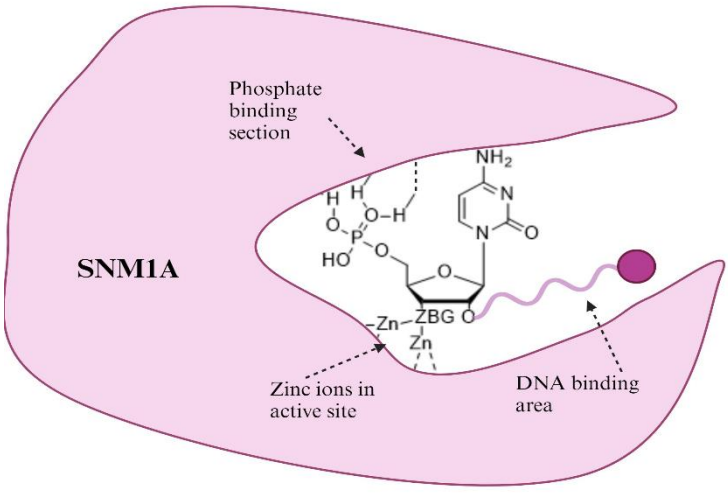


Figure 5: Adhesion of the proposed active site of the enzyme to DNA[41]

The McGouran group have carried out research into the development of substrate-mimic inhibitors for SNM1A.[26,29,31 34] Modified DNA oligonucleotide-bearing inhibitors at the 5'-end **(1, 2, 3 Fig. 6)** were shown to inhibit SNM1A. Nucleoside derivatives with a malonate-based ZBG at the 5' position **(**inhibitor **1, Fig. 6.)** were shown to inhibit SNM1A at a 1 mM concentration.[29] Similarly squaramide moieties (inhibitor 2, **Fig.6.)** act as inhibitors as they have a strong preference for chelating cations such as zinc through their two carbonyl oxygen atoms, therefore allowing for inhibition of the enzyme.[33] Triazole-modified linkages are a rigid mimetic of an azide linkage,[32,34] therefore affording the inhibitor high chemical stability leading to subsequent resistance to hydrolysis. Triazole linkages are biocompatible as they are easily transcribed and replicated in cells.

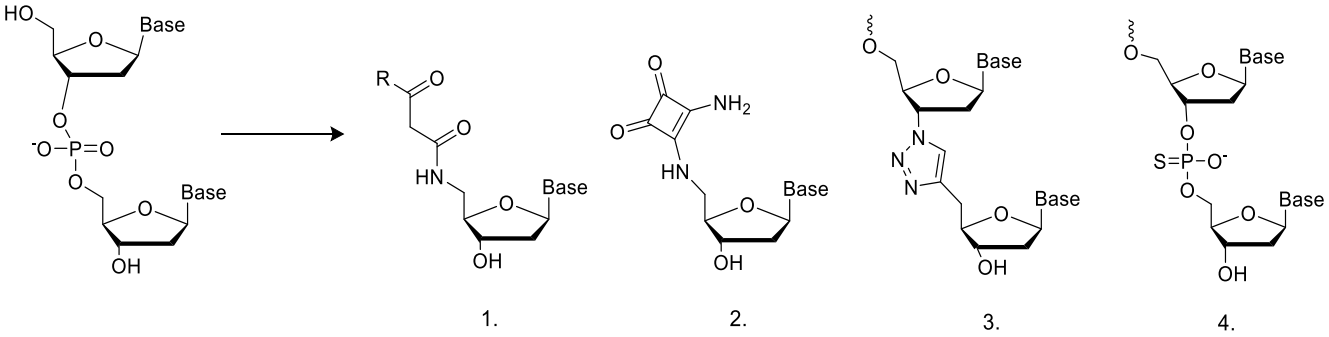


Figure 6: 1. Malonate inhibitor, 2. Squaramide inhibitor, 3. Triazole inhibitor 4. Thiosulfate inhibitor

The induction of phosphorothioate (PS) bonds (inhibitor 4, **Fig 6.)** to create an inhibitor has been reported by Eckstein,[39] which involves removing one of the bridging oxygen atoms that make up the phosphate backbone with a sulphur atom. These are useful as they increase stability and prevent the incidence of nuclease degradation.

1.5 Oligonucleotide synthesis and modification

Oligonucleotides are mainly used for modulating gene and protein expression and are increasingly being utilized in diagnostics, therapeutics and probes due to their greater target binding affinity. The natural binding that exists between nucleotides is through a 3'-5' phosphodiester linkage.[42]The phosphodiester linkage allows for structural stability and

preservation of the DNA structure. The modification of the 2'-5' linkage **(Fig. 7)** has been investigated. Deoxyribose one of the main constituents of DNA does not have an available 2' hydroxyl group where a phosphodiester bond could form. However, RNA does possess an available 2'-5' linked group that can form base pairs with complementary RNA. The majority of RNAs have 3'-5' linkages yet the 2'-5' linkage also exists in nature and has roles in viral replication and mRNA processing.[52] This linked group however has poor thermal stability in contrast with the 3'-5' linkage.[49,50] This thermal instability is thought to be associated with the base pair stacking that can take place with 2'-5' linked compounds[49,51] due to irregular pairing between bases such as guanine and uridine.[51] Certain combinations of base pairs form more stable interactions than others, so nearest neighbour base-stacking interactions are important determinants of DNA stability. The 2'-5' linkage is proven to be resistant to enzymatic hydrolysis as the spatial arrangement and positioning of the atoms with this linkage create steric hinderance preventing potential incidence of hydrolysis

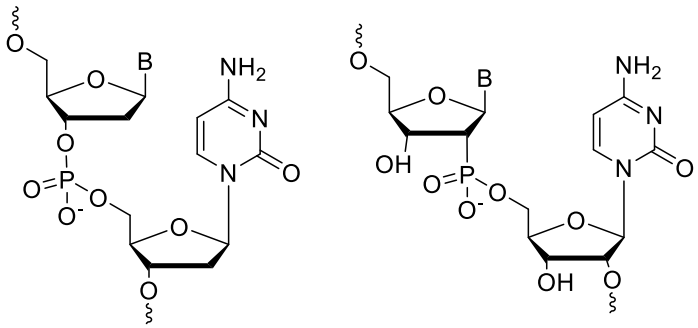


Figure 7: Natural 3'-5' linkage (left) and modified 2'-5' linkage (right)

1.6 Activity based Probes

Activity based probes (ABPs) are small molecules composed of a reactive warhead, a linker and a tag which allow for specific labelling and enrichment of target proteins. ABPs can analyse the regulation of protein activity that would not be available by traditional proteomic methodologies, as they can be captured and studied in their active form.[31] The 'warhead', which is the reactive functional group, covalently binds to the catalytic residue of the protease active

site. The warhead sets the precedent as its charge, polarity and reactivity determine the specificity of the probe[30]. Meanwhile, the linker separates the functional group from the tag, whilst maintaining a chain like connection with minimal steric hindrance between both groups **(Fig.8)**. Incorporation of a tag group into an activity-based probe enables the visualization and/or purification of labelled proteins. Designing an activity-based probe for SNM1A allows for the study of its activity which is relatively unknown.

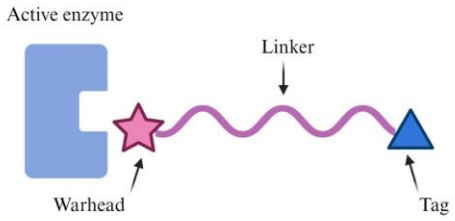


Figure 8: Activity based probe structure.[42]

1.7 Aims and Objectives

The purpose of this project was to synthesise a modified mononucleoside to study SNM1A, that could be manipulated and inserted into an oligonucleotide to act as an activity-based probe. Rather than digest SNM1A, the modified nucleotide will capture the enzyme allowing for further study. The target synthetic scheme included below shows the proposed nucleoside that will be incorporated into the oligonucleotide. As discussed earlier, the natural phosphate linkage present in the oligonucleotide is vulnerable to hydrolysis.[34] Therefore this linkage is swapped by the insertion of a zinc binding group to allow the probe to be recognised by the enzyme whilst being different enough to evade hydrolysation by the enzyme. As shown in the scheme below the natural 3'-5' linkage is replaced by the hydrolysis resistant 2'-5' linkage.

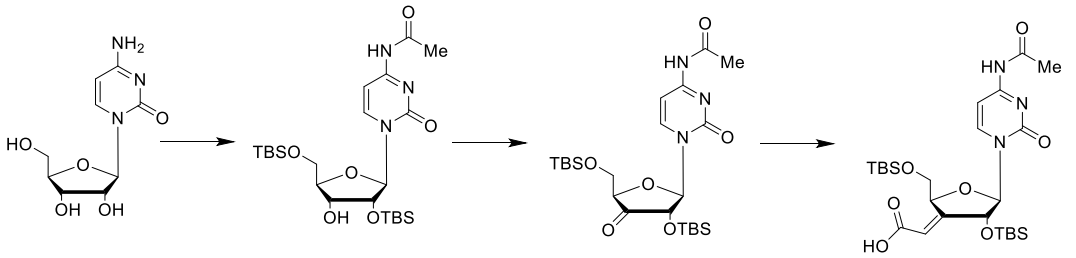


Figure 9: Target synthetic scheme.

Cytidine was utilised as a starting material to create the oligonucleotide. The subsequent mononucleosides created would be attached to the oligonucleotide strand, which would be carried out by means of oligonucleotide solid state synthesis. This is because artificial backbones are typically introduced by modified on-resin coupling of a monomer in order to form the unnatural backbone or via coupling a dinucleotide that contains an artificial linkage. Once the synthesis is complete, the resulting oligonucleotide would be tested against SNM1A to evaluate the potential it has to bind to the enzyme and inhibit its activity. Cell assays will be run in order to assess the DNA damage repair activity and see if it can be used to inhibit ICL repair in chemotherapeutics.

2. Results & Discussion

2.1 Synthesis Strategy of Target Compounds and Protecting Group Implementation

The aim of this project was to synthesise an advanced intermediate **5** in order to access modified cytidine derivatives **15-17** (**Fig. 10)**. Successful use of cytidine analogues such as the HIV antiviral Zalcitabine **18** which utilises a 2',3'-dideoxyribose scaffold, allows for consideration of other potential modifications to the 3' position. A cytidine derivative with a modification performed at the 3' position would be a useful target for zinc binding groups as evidenced by previous work carried out by the McGouran group,[26,29,33,34] where squaramide, malonate and triazole groups were successfully demonstrated to be effective inhibitors of SNM1A.

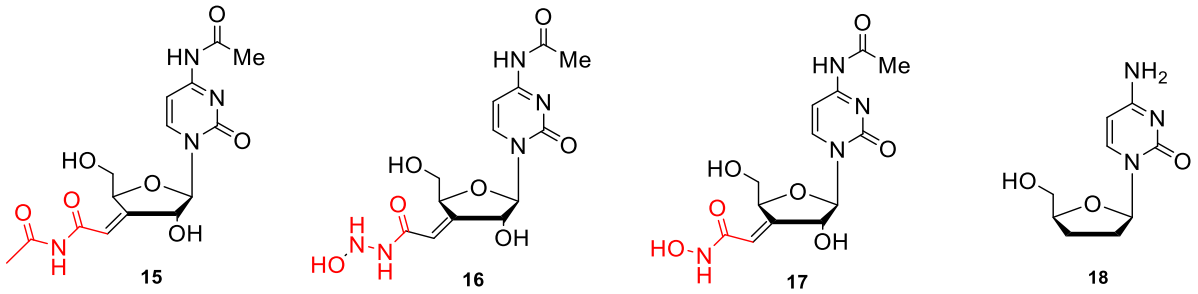


Figure 10: Target nucleotides **15**,**16**,**17** and modified cytidine analogue Zalcitabine **18 (ZBG shown in red)**

Nucleosides **15**, **16** and **17** were the initial target inhibitors with a modified 3' position, these nucleosides can either be used by themselves or can be incorporated into an oligonucleotide strand and tested against SNM1A. As such a retrosynthetic analysis **(Fig. 11)** was performed in order to devise a suitable synthetic strategy.

When considering the forward synthesis of this retrosynthetic analysis shown in (**Fig. 11**) it is necessary to protect the 2'- and 5'- hydroxyl groups so the 3'- position can be selectively modified.

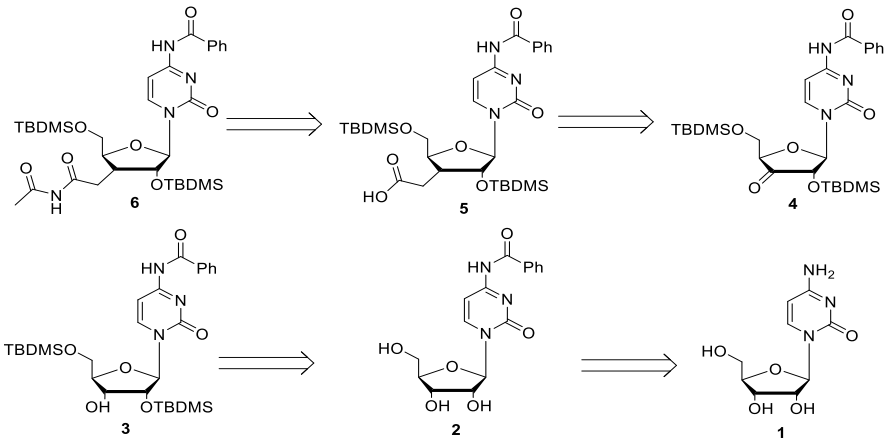


Figure 11: Retrosynthetic analysis of the target compound **6.**

Protecting groups are a vital aspect in the synthesis of these target molecules as they are used to prevent unwanted side reactions with other functional groups. *tert*-butyldimethylsilyl chloride (TBDMSCl) reacts selectively with the 5'-hydroxyl groups but can also protect the 2'-position in deoxynucleosides.[40] As such, this protecting group is often preferential in the synthesis of modified nucleosides. In addition, the removal of TBDMS is a facile process when using acidic conditions such as HCl or fluoride sources such as TBAF, or HF. Furthermore, TBDMS is stable to oxidation mediated by both (2,2,6,6-tetramethylpiperidin-1-yl) 1-oxyl (TEMPO) and (diacetoxyiodo)benzene (BAIB) and Dess Martin Periodinane (DMP). Therefore, the TBDMS group was the optimal choice for protecting the 2'- and 5'- hydroxyl groups.

The functional group interconversion from an alcohol to a ketone at the 3' position would allow for incorporation of the zinc binding group. In previous strategies to access similar modified nucleosides, protection of the exocyclic amine (N4 position) is essential for prevention of unwanted side reactions. Acetyl, benzoyl and isobutyl groups have been commonly utilised to protect amino functional groups during nucleoside synthesis. [47]

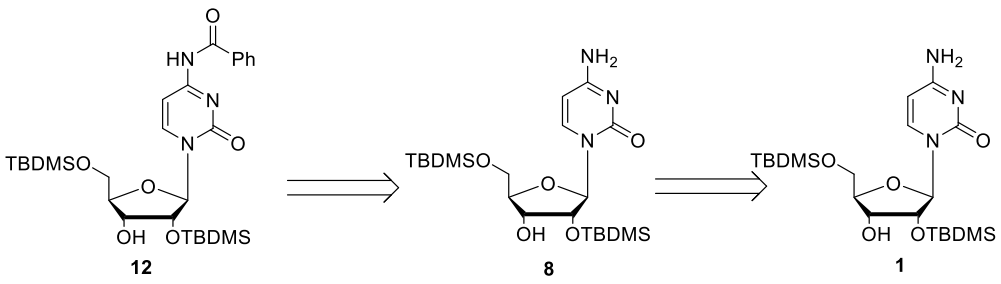
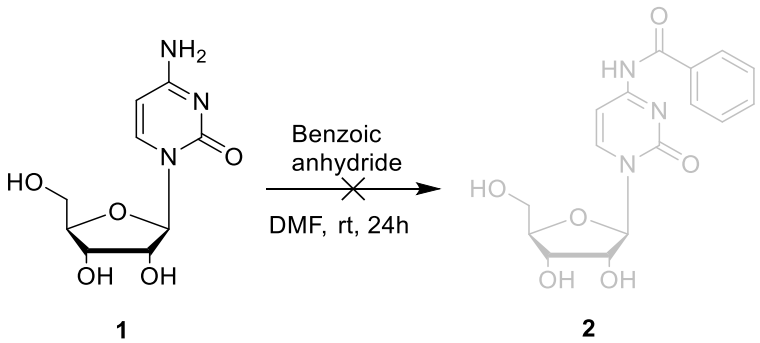


Figure 12: Retrosynthetic analysis of benzoylated nucleoside **12** at N4

An alternate protecting group strategy **(Fig. 12)** was devised involving the incorporation of the benzoyl (Bz) protecting group, to give **12**. This protecting group is commonly used in oligonucleotide synthesis for the protection of the N4 position. Orthogonal protection is used in this synthesis as a result of multiple functional groups present on the nucleoside. The N4 group in this case lies orthogonal to the TBS groups. Using the orthogonal protection strategy will allow for selective deprotection of the amine leaving the other protecting groups unaffected.

2.2. Benzoylation of the Exocyclic Amine

Initial benzyl protection of the exocyclic amine was attempted using benzoic anhydride, (**Scheme 1).** It was decided to protect the N4 amine first instead of the 2'- and 5'- hydroxyls due the basic and reactive nature of amines.



Scheme 1: Reaction of Cytidine with Benzoic Anhydride

Entry	Protecting Reagent	Temperature	Time	Solvent
1	Benzoic (Bz) Aanhydride	rt	24h	Anhydrous DCM
2	Bz anhydride	rt	24h	DMF
3	Bz anhydride	rt	36h	Anhydrous DMF
4	Bz anhydride	rt to 30°C	48h	DMF
5	Bz anhydride	rt	72h	DMF
6	Bz anhydride	rt to 30°C	72h	DMF

Table 1: Reaction conditions for attempted formation of **7** experiments (1-6)

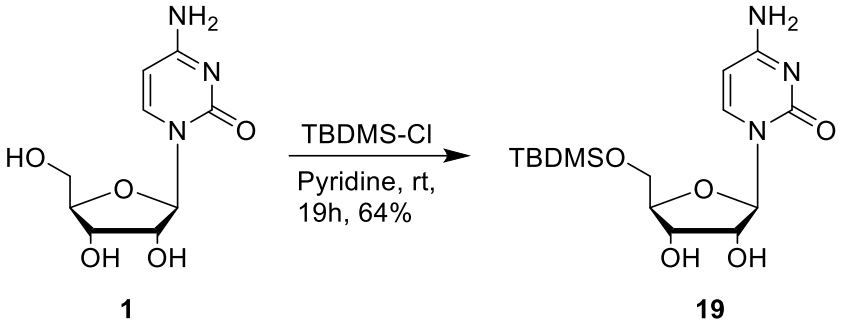
Benzoic anhydride was reacted with cytidine in DCM for 24 h (**Table 1, Entry 1).** After this time TLC analysis (EtOAc: Hexane 3:7) showed no consumption of the starting material at (R_f=0.4) and no formation of product. Further analysis of this reaction by ^{1}H NMR confirmed that the starting material had not reacted. Compound **1** had poor solubility in DCM, forming a suspension, and as such the reaction was repeated in DMF (**Table 1, Entry 2)** which had considerably improved solubility, however this also yielded unreacted starting material.

Variation of reaction time was tested with the reaction proceeding for 36, 48 and 72 hours, **(Entries 3 -6 respectively)**. Variation of temperature was also tested in order to aid with solubility of the materials in DMF. After 72 hours with gradual heating to 30°C **(Table 1, Entry 6)**, two separate TLC spots appeared with the starting material cytidine at (R_f = 0.3) and the product at (R_f = 0.6). Purification of the product was attempted using flash chromatography (Hex: EtAOc 95:5-80:20) resulting in the product being obtained as a mixture. Analysis of the ^{1}H NMR spectrum indicated cytidine and the product present in a 1:3 ratio. Further purification was attempted by dissolving **7** in a 1:1 mixture of EtOAc and MeOH to attempt recrystallisation

of the product. Neither compound was soluble in this solvent system and remained as a suspension. Gradual heating of the solvent to 30℃ also failed to improve solubility. A plausible explanation for the lack of reactivity could be as a result of benzoic anhydrides moisture sensitivity and as such when stored for long periods of time can hydrolyse to the less reactive benzoic acid.

2.2.1 2'- and 5'- Protection and Oxidation

After difficulties experienced protecting the N4 position on **1** it was decided to instead attempt protection of the 2'- and 5'- hydroxyl group positions to obtain **8 (Scheme 3)**. TBDMS was chosen as the protecting group as it can be used to protect both of these positions at the same time. The first attempt to synthesise this compound led to monosubstitution occurring at the 5'- position (**Scheme 2)**. This was expected as TBDMS-Cl reacts preferentially with the 5'-position in deoxynucleotides due to the enhanced nucleophilicity of the primary alcohol.[39]

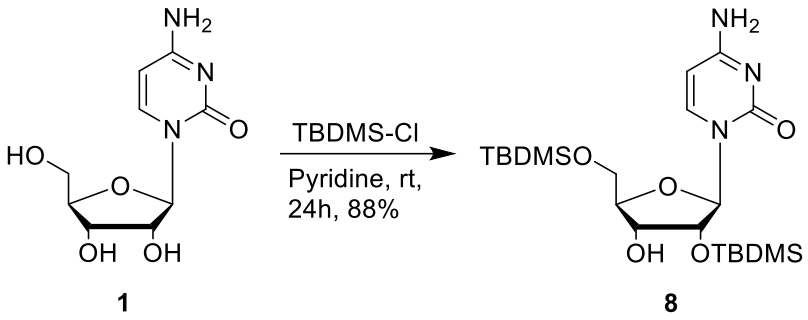


1 **19**

Scheme 2: Addition of OTBS groups to the 5' position

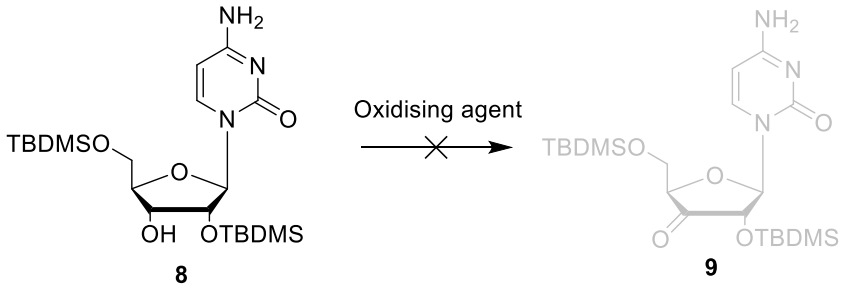
TBDMS-Cl was added to pyridine under argon and left to react with compound **1** for 19 hours. This reaction led to protection of only one of the alcohol groups at the 5' hydroxyl position to give **19** with a 64% yield**.** Whilst monoprotection of the 5' position indicated the reaction went to completion; it did not lead to formation of the desired compound. Restricting the reaction time to 19 hours and using only 3 equivalents of TBDMS-Cl ensured that only the 5'-hydroxyl reacted. A longer reaction time of 24 hours and adding 2 additional equivalents of TBDMS-Cl

provided the desired compound **8** in a good yield of 76%. Whilst a good yield was recorded, a significant amount of the modified compound **8** was required in order to form more than one modified mononucleotide and to attempt different oxidation methods. The reaction was repeated again to a larger scale resulting in a yield of **8** (1.91g, 88%).



Scheme 3: Addition of OTBS groups to the 2'- and 5'- positions.

Oxidation of the 3' alcohol was then attempted but proved to be challenging. This is likely as a result of the unprotected amine group at N4. Dess Martin Periodinane (DMP) which is a hypervalent iodine compound, was utilised due to its mild oxidation conditions and mild purification method (**Table 2 Entry 1).**



Scheme 4: General oxidation scheme

Initial dissolution of DMP in anhydrous DCM proved slow due to solubility constraints. Once the reaction mixture was allowed to warm to room temperature, DMP was partially soluble. TLC analysis (DCM: MeOH, 90:10) after 3 hours showed presence of starting material

indicating that no reaction took place. Additional reaction time did not result in the reaction progressing. Pyridine was utilised as a basic buffer to prevent the substrate being exposed to the acetic acid by-products of the reaction whilst lowering activation energy and increasing alcohol oxidation. It can be theorised that the unprotected N4 group interfered with the oxidation of the secondary alcohol and underwent side reactions.

In a study by Roberts *et al*, [47] they propose that steric hinderance can prevent effective oxidisation in nucleosides leading to no reaction taking place.

Entry	**Oxidising agent**	**Solvent**	**Reaction time**	**Temperature**
1	DMP, pyridine	Anhydrous DCM	4 h	0℃ to rt
2	DMP, pyridine	Anhydrous DMF	3 h	0℃ to rt
3	DMP, pyridine	Anhydrous DMF	4 h	0℃ for 30 minutes and gradual heating to rt
4	TEMPO/BAIB	CH_3CN-H_2O	5 h	rt

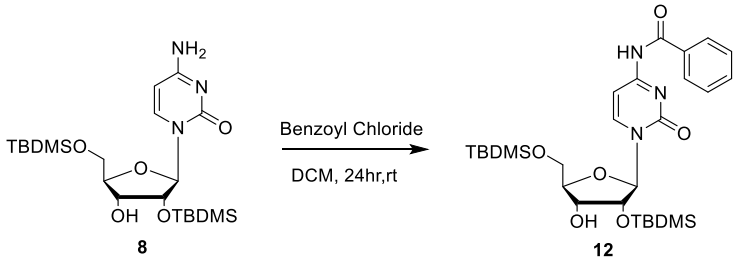
Table 2: Oxidisation reactions attempted to form **3**

Oxidation of the 3'- position was repeated using anhydrous DMF **(Table 2 Entry 2)** instead of anhydrous DCM due to improved solubility of DMP in DMF, this reaction did appear to go to completion after 3 hours. Whilst formation of a new product was confirmed via TLC (R_f=0.6), purification via liquid-liquid extraction proved difficult and led to impurities remaining in the product. Large quantities of brine, EtOAc and $NaHCO_3$ were used to carry out extraction. TLC analysis indicated that the majority of the assumed product went into the aqueous phase. ^{13}C NMR analysis of both phases indicated that the reaction did not go to completion as no product was isolated despite initial indications from TLC analysis.

DMP oxidation was tested once more by reducing the reaction temperature to 0°C for 30 minutes instead of 15 minutes as tested previously (**Table 2 Entry 3**), however yet this also proved unsuccessful so other oxidation methods were investigated. TEMPO/BAIB mediated oxidation was utilised in another attempt to oxidise the secondary alcohol and form the desired compound **9, (Table 2 Entry 4)**. TEMPO/BAIB was tested due to its mild oxidation conditions and compatibility with TBS protecting groups. This reaction was carried out for 5 hours with TLC analysis (9:1 DCM: MeOH) showing unreacted starting material (R_f=0.4). A plausible reason for the unsuccessful oxidation attempts of the 3' position could be as the N4 amine group was left unprotected. Therefore, it could participate in the reaction with TEMPO/BAIB preventing oxidation from occurring. Whilst protection of the 2' and 5' positions proved successful the synthesis could not move any further without addressing the protection of the amine functional group.

2.2.2 Protection of the Exocyclic Amine

Whilst initial benzoylation of the N4 position of cytidine was unsuccessful, it was decided to attempt benzoyl protection of TBDMS protected **8** using benzoic chloride in the presence of pyridine. TLC analysis (DCM: MeOH, 92:8) indicated formation of the product and complete consumption of the starting material. However, **12** was not isolable via flash column chromatography. ^{1}H NMR analysis was performed suggesting degradation of the compound **12** with starting material **8** present in a 1:2 ratio.

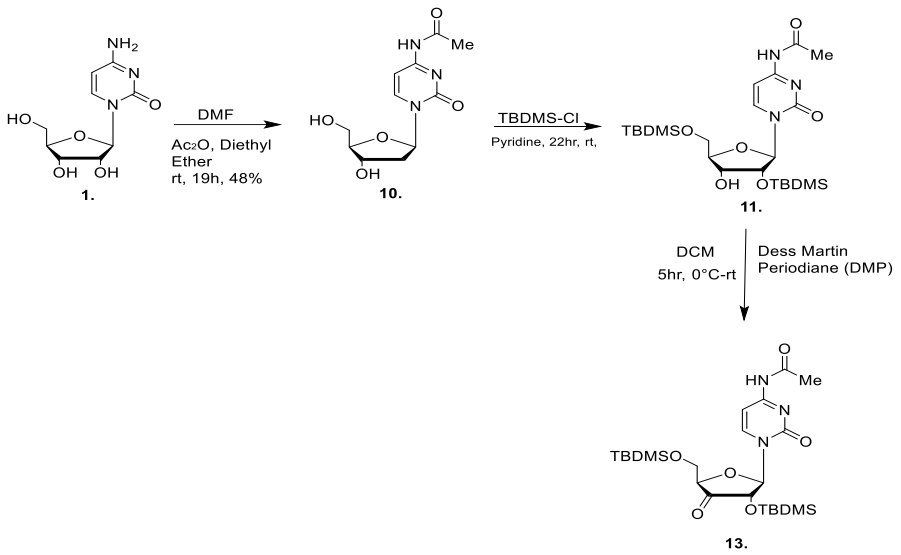


Scheme 5: Synthesis of N4 position using benzoyl chloride

While benzoylation of the N4 position was confirmed via ^{1}H NMR, it led to subsequent deprotection of the TBDMS groups at the 2' and 5' position. This was unexpected as generally TBDMS groups are stable unless exposed to highly acidic conditions or fluoride sources. A reason for this could involve the slight acidity of silica used during flash chromatography leading to degradation of the TBDMS groups.

2.3 Revised synthetic strategy

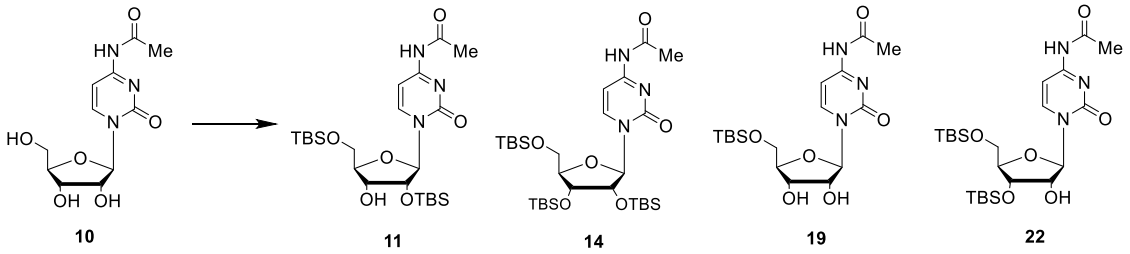
As a result of the synthetic challenges encountered accessing suitably protected cytidine derivatives and subsequent oxidation, a second synthetic pathway was derived, starting with acetylation of the N4 position, (**Scheme 6)**



Scheme 6: Synthetic scheme involving implementation of Ac group

Acetylation of the N4 position on **1** was carried out with acetic anhydride at room temperature to obtain the desired compound **10** with a moderate yield (48%). Acetyl groups are orthogonal to the TBDMS groups and as such these protecting groups were utilised again for selective modification of the 2' and 5' positions to give compound **11**. However, degradation was observed during attempted purification. Crude analysis of the reaction mixture using ^{1}H NMR and mass spectrometry indicated the presence of compound **11**, however, the desired

compound could not be isolated as one unprotected alcohol. It is possible that exchange takes place between the 2' and 3' position (**Fig.11)**. Silica and its slight acidity will only amplify this trait as mentioned previously. Conversion of silyl groups in close proximity to alcohols can be achieved under mildly acidic conditions.[36] Instance of protection at the 2' and 5' position is more likely than at the 3' position, however it is hypothesised that interference occurs between the TBS group and the unprotected alcohol, leading to migration between these positions. A complex mixture of products was seen during ^{1}H NMR analysis with the main constituents being **11** and **22**. Due to constant degradation the exact ratio these compounds were present in could not be determined.



10 11 14 19 22

Figure 11: Possible TBS protection positions

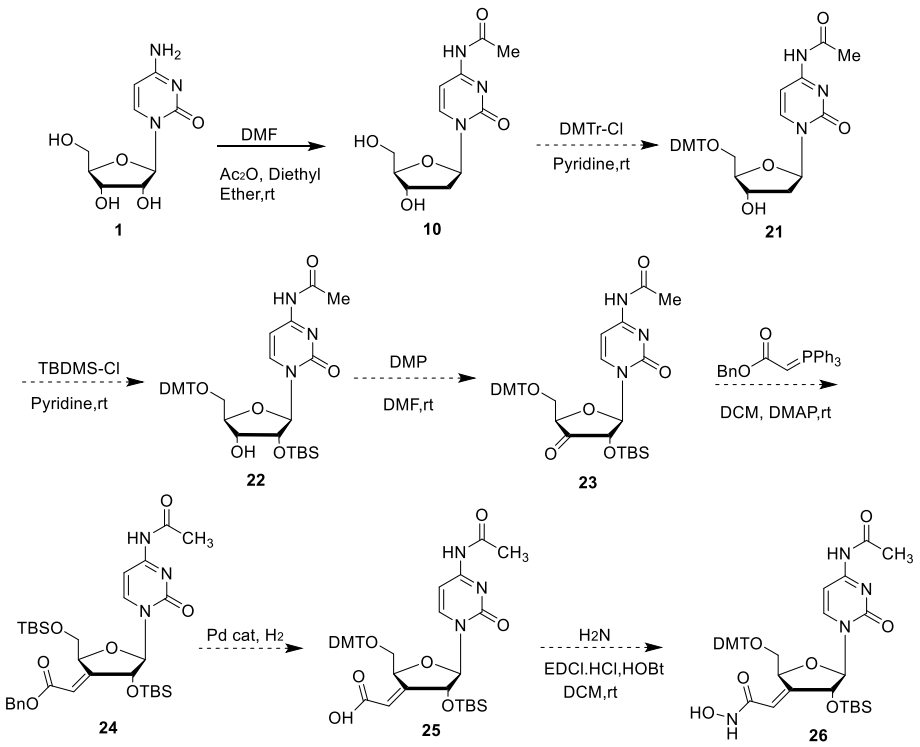
Despite difficulty identifying the TBS protected groups as shown in (**Fig 11),** oxidation with DMP was attempted in order to access **13.** Due to time constraints flash chromatography could not be performed in order to purify the compound. However, ^{13}C NMR suggested possible formation of the product due to emergence of a C=O peak at 172.4 ppm **(Figure 13 (bottom)**.



Figure 13: ^{13}C NMR comparison of **11** (top) and **13** (bottom).

3. Conclusions and Future Work

In conclusion, cytidine was successfully protected at the 2' and 5' positions using the TBDMS group, however challenges were encountered when attempting the subsequent oxidation of the 3' position using DMP and TEMPO/BAIB. As a result of multiple unsuccessful attempts at oxidation using different methods, a second protecting group strategy was derived using the acetyl protecting group. Through this alternative synthetic pathway, successful acetylation of the exocyclic amine was undertaken, followed by attempted protection of the 2' and 5' hydroxyl groups using TBDMS, however purification proved challenging and isolation of the target molecule elusive. Oxidation was attempted with DMP but due to time constraints purification of **13** was not possible however analysis with NMR spectroscopy indicates the high possibility of presence of desired product.



Scheme 7: Future Synthetic Steps

Due to the difficulty in purification of the compounds with multiple TBDMS groups, future work on this synthetic route should involve an alternative protecting group strategy (**Scheme 7)**, Following acetylation of the exocyclic amine, 5' protection could be carried out with DMTr-Cl. 2' protection could be undertaken with TBDMS-Cl to allow for selective oxidation of the 3' position. This suggested protecting group strategy would enable 3' modification to incorporate metal binding groups. Modification of the carboxylic acid group on **25** would allow for mononucleoside **26** which can be incorporated into an oligonucleotide through solid phase synthesis.

The activity of SNM1A and the strength of its inhibition ability can be evaluated by running several gel-based assays. Incubating SNM1A with the modified oligonucleotide substrate would hopefully leads to digestion of the substrate from the 5'-end, one nucleotide at a time. If these assays yielded positive results, it would be possible to screen other ZBG to test their inhibition ability.

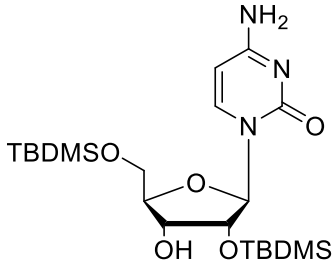
4. Experimental Methods

General Methodology

During the course of the project, chemicals were purchased from Fluorochem, Thermo Fisher Scientific, Merck and Sigma Aldrich. These chemicals were used without need for further purification. THF was dried over molecular sieves under argon. Oxygen-free anhydrous argon was acquired from BOC gases. Flash column chromatography was performed using silica gel purchased from Sigma Aldrich (particle size ranged from 0.04-0.06 mm). Thin layer chromatography (TLC) analysis was executed by using pre-coated thin layer silica gel 60 F_{254} sheets. These plates were exposed to UV irradiation (254 nm) to visualise formation of products. Anisaldehyde (9.2 mL p-methoxybenzaldehyde, 3.75 mL Acetic Acid, 336 mL 95% EtOH, 12.75 mL conc. H_2SO_4) and ninhydrin (1.5 g ninhydrin, 5 mL Acetic Acid, 500 mL 95% EtOH), were utilized to stain the TLC plates. NMR spectra were recorded using Bruker 400MHz and 600MHz spectrometers in DMSO-d_6. NMR spectra were assigned by utilising ^{1}H, ^{13}C, HSQC, HMBC, NH COSY and selective ^{1}D TOCSY experiments. The coupling constants in this experiment were recorded in Hertz (Hz) accurate to 0.2 Hz whilst chemical shifts (δ) were reported in ppm. The nucleosides included in this report adhere to standard nucleoside convention. Mass spectroscopy measurements were taken on a Bruker ESI. Infrared spectra (IR) were taken on a Perkin Elmer spectrophotometer. Melting points of compounds were determined using a Griffin melting point apparatus.

Experiments highlighted in grey were attempted and proven to be unsuccessful. General methodology is outlined in the title of the experiment as several of the experiments were tested a number of times with slight adjustment of variables each time but followed the same general procedure.

2',5'-Bis-*O*-(*tert*-butyldimethylsilyl) cytidine



8

Cytidine (1.46 g, 6.02 mmol) was dissolved in pyridine (14 mL), under argon, at room temperature. *t*-Butyldimethylsilyl chloride (TBDMS-Cl) (3 g, 19.90 mmol) was added, and the reaction mixture was stirred for 19 hours, under argon. TLC analysis was performed (DCM: MeOH 85:15), showing incomplete consumption of the starting material (R_f = 0.1) and the formation of the product (R_f = 0.7). To afford completion, two more equivalents of TBDMS-Cl (266mg,1.76mmol) was added and the mixture was stirred for a further 3 hours. The solvent remaining was removed *in vacuo* and the product was purified via flash chromatography (DCM: MeOH, 95:5-92:8). The product was obtained as an orange powder (1.91g, 88%).

^{1}H NMR (600 MHz, DMSO-d$_6$): δ = 0.07 (s, 2H, CH_3^{TBDMS}), 0.08 (s, 4H, CH_3^{TBDMS}), 0.10 (s, 6H, CH_3^{TBDMS}), 0.86 (s,7H, *t*-Bu^{TBDMS}), 0.89 (s, 3H, *t*-Bu^{TBDMS}),0.92(s,8H, *t*-Bu^{TBDMS}), 3.75 (d, $J_{5',a,5'b}$ = 11.4 Hz 1H, H-5'a), 3.83 (m, J = 4.2 Hz, 1H, H-2'), 3.93 (m, 2H, H-3', H-4') 4.95 (d, $_{J3',OH-3'}$ = 4.8 Hz, 1H, OH-3'), 5.67 (d, $J_{5,6}$ = 7.8Hz, 1H, H-5), 5.80 (d, $J_{1'-2'}$ = 3.7 Hz, 1H, H-1'), 7.13 (d, J=9.1 Hz , 2H, NH_2), 7.85 (d,1H, $J_{5,6}$ = 7.5 Hz, H-6) ppm.

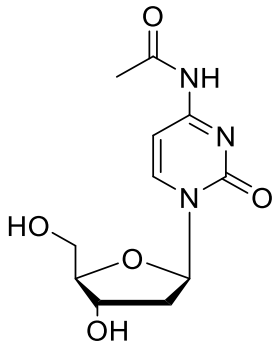
^{13}C NMR, (151 MHz, DMSO-d$_6$): δ = -5.1 (CH_3 ^{TBDMS} x 2), -4.6 (CH_3 ^{TBDMS}), -4.5(CH_3 ^{TBDMS}), -4.4 (CH_3 ^{TBDMS}), 18.4 (qC t-^{BuTBDMS}), 18.5 (qC t-^{BuTBDMS}), 26.1(*t*-Bu^{TBDMS}), 26.2 (*t*-Bu^{TBDMS}), 62.5 (C-5'),69.3(C-4'), 76.9 (C-2'), 83.8 (C-3'), 94.0 (C-5), 140.9 (C-6), 155.2 (C-2) and 165.7 (C-4) ppm.

ν_{max}/cm^{-1} = 3224 (N-H), 3163 (N-H), 2985 (O-H), 2840 (C-H),1651 (C=O), 1638 (N-H), 1614 (C=C), 1555 (N-H), 1488 (C-N, C=N), 1441 (C-N), 1397 (C=CH), 1305 (C-N), 1223 (C-N), 1192 (C-O), 1129 (C-O), 1069 (N-C-H), 1019(C-C-H), 966 (C-C), 701 (N-H).

m.p: 192-193 °C (lit [43]: 198-201 °C).

HRMS (APCI+): *m/z* calc: 472.2658, found 472.2657

N4-Acetyl-2'-deoxycytidine



10

2'-Deoxycytidine 18 (2.00 g, 8.81 mmol) was suspended in anhydrous DMF (44 mL), under argon. Acetic anhydride (0.92 mL, 948 mg, 9.28 mmol) was added and the solution was stirred at room temperature, for 32 hours. After this time, TLC analysis showed complete consumption of the starting material (R_f = 0.3, DCM-MeOH, 1:1) to give the product (R_f = 0.5, EtOAc: MeOH, 4:1). The solvent was removed *in vacuo* and DMF was removed via co-evaporation with toluene (10x10 mL). Diethyl ether (5 mL) was added to the residue, resulting in the precipitation of a white solid. The supernatant was removed and the solid was further washed with diethyl ether (3.5 mL). The resulting solid was purified via flash chromatography (DCM: MeOH 96:4 to 88:12). The product was dried under high vacuum and a yield of (957 mg, 48%) was reported.

^{1}H NMR (600 MHz, DMSO-d$_6$): δ= 2.10 (s, 3H, CH_3^{Ac}), 3.60 (m. 2H, H-5'), 3.74 (m, 1H, H-4'), 3.90 (dt *J* = 5.9, 3.0 Hz, 1H,H-3'), 3.98 (m, 2H, H-2'), 5.07 (d, $J_{3',OH}$ = 5.3 Hz, 1H, 5'OH),

5.15 (t, $J_{3'OH}$= 5.3 Hz, 1H, 3'OH), 5.48 (d, *J* = 4.7 Hz, 1H, H-1'),), 5.78 (d, *J* = 3.0 Hz, 1H, H'3), 7.18 (d, $J_{5,6}$= 7.5 Hz, 1H, H-5), 8.42 (d, $J_{5,6}$ = 7.5 Hz, 1H, H-6), 10.87 (s, 1H, NH-4) ppm.

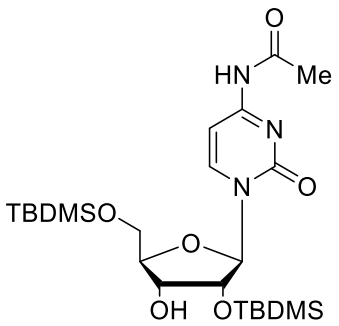
^{13}C NMR, (151 MHz, DMSO-d$_6$): δ = 24.8 (CH_3^{Ac}),60.4 (C-2'),69.0 (C-5'), 74.9 (C-3'), 84.6 (C-4'), 90.6 (C-1'), 95.6 (C-5), 145.8 (C-6), 155.1 (C-2), 162.6 (C-4) and 171.4(CO^{Ac}) ppm.

ν_{max}/cm^{-1}: 3224 (N-H), 3163 (N-H), 2985 (O-H), 2840 (C-H),1651 (C=O), 1638 (N-H), 1614 (C=C), 1555 (N-H), 1488 (C-N, C=N), 1441 (C-N), 1397 (C=CH), 1305 (C-N), 1223 (C-N), 1192 (C-O), 1129 (C-O), 1069 (N-C-H), 1019(C-C-H), 966 (C-C), 701 (N-H).

m.p: 135℃ (lit.[44] 150-170℃)

HRMS (APCI$^+$): *m/z* calc: 292.0904 [M+ Na]$^+$, found 290.0785

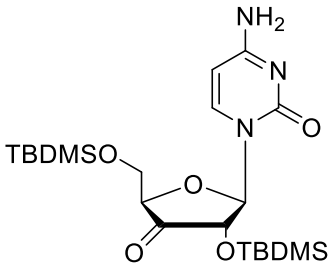
N4-Acetyl-2',5'-Bis-O-(*tert*-butyldimethylsilyl) cytidine



N4-Acetyl-2'-deoxycytidine (350 mg, 1.21 mmol) was dissolved in pyridine (3.5mL), under argon, at room temperature. *t*-Butyldimethylsilyl chloride (TBDSCl) (665 mg, 19.90 mmol) was added, and the reaction mixture was stirred for 22 hours, under argon. TLC analysis was performed (DCM: MeOH 9:1), showing complete consumption of the starting material (R_f = 0.1) and the formation of the product (R_f = 0.5). The solvent remaining was removed *in vacuo* via co-evaporation with toluene. Purification was attempted via flash column chromatography

(DCM: MeOH, 95:5-93:7) but this led to degradation of the product. Some of the desired product was collected as a mixture and used without further purification. ^{1}H NMR is not included as a result of a complex mixture of products remaining in solution. ^{13}C NMR was utilised as a tool to see whether protection had taken place and is included in the appendix.

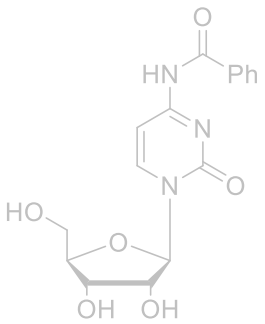
2',5'-Bis-O-(*tert*-butyldimethylsilyl) cytidine (attempted)



9.

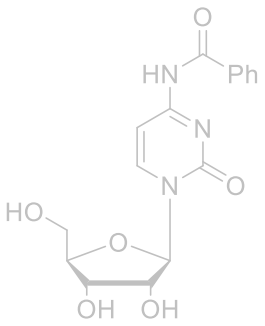
2',5'-Bis-O-(*tert*-butyldimethylsilyl) cytidine (90 mg, 0.411 mmol) was dissolved in anhydrous DCM (1 mL). Separately, Dess Martin Periodinane (925 mg, 0.0065 mol) was dissolved in anhydrous DCM, Initially, (7 mL) of anhydrous DCM was added dropwise, but a further 1mL was added to ensure the full dissolution of the DMP and was added to the reaction mixture at 0°C. This mixture was stirred under argon at 0° C for 15 minutes and then at room temperature for a further two hours. TLC analysis was carried out using (8.5:1.5 DCM: MeOH). Full consumption of the starting material at (R_f = 0.3) was observed, alongside emergence of the product at (R_f = 0.4). Due to time constraints flash column chromatography was not carried out to enable further purification.

N4-Benzoyl-2'-deoxycytidine (General Procedure)



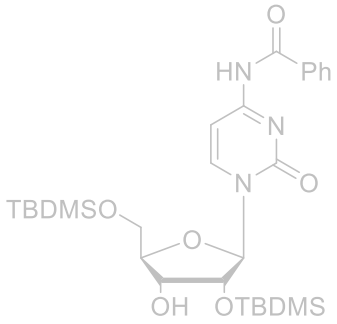
Benzoic Anhydride (102 mg,0.45 mmol) was added to cytidine (100 mg,0.41 mmol) in DMF (3 mL). An inert environment was enabled keeping the reaction under argon. The reaction mixture took a few seconds to dissolve in DMF. The reaction was left for 24 hours. TLC analysis (EtOAc: Hexane 3:7) indicated starting material still present at (R_f = 0.4) and product spot at (R_f = 0.8). Remaining solvent was removed via co-evaporation with toluene (5x5 mL) The product was quenched using H_2O (10 mL). It was washed using (3x10 mL) EtAOc. The organic layer was washed again with brine (10 x 10 mL). TLC analysis (EtOAc: Hexane 3:7) indicated that the starting product remained in the organic layer (R_f = 0.4). The organic layer was filtered and dried over $MgSO_4$. Flash column chromatography was carried out to further purify the product (Hex: EtAOc 95:5-80:20). This did not yield the desired product. This reaction afforded recovered starting material.

N4-Benzoyl-2'-deoxycytidine -2 (attempted)



Benzoic Anhydride (302 mg,0.452 mol) was added to cytidine (300 mg,0.411 mmol) in DMF (9 mL). An inert environment was enabled keeping the reaction under argon. The reaction mixture was stirred for 20 hours at room temperature. DMF was removed in vacuo and the residue was triturated with diethyl ether (2 x 100 mL). The resulting white solid was collected via suction filtration and washed again with diethyl ether (2 x 5 mL). TLC analysis (DCM: MeOH 9:1) indicated that starting material remained (R_f = 0.1) alongside formation of the product (R_f = 0.6). Recrystallisation with EtOAc: MeOH, 2:3 did not yield the desired product. Recrystallisation was attempted once more with isopropanol but was unsuccessful.

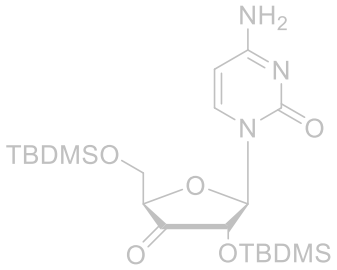
N4-Benzoyl-2',5'-bis-O-(*tert*-Butyldimethylsilyl)-3'-oxocytidine (attempted)



2',5'-Bis-O-(*tert*-butyldimethylsilyl) cytidine **(8)** (300 mg, 0.635 mmol) was dissolved in anhydrous pyridine (4 mL), under argon. Benzoyl chloride (1.5 mL) was added dropwise to the solution and left to stir overnight at room temperature. TLC analysis (DCM: MeOH, 9:1)

showed consumption of the starting material (R_f = 0.3) and the formation of the product (R_f= 0.7) along with another spot (R_f=0.9). The solvent was removed under reduced pressure and by washing the reaction mixture with toluene (3x10 mL). The resulting residue was purified via flash chromatography (Hex: EtOAc 93:7 to 70:30, H_2O: isopropanol: EtOAc 1:4:5). This did not yield the desired compound.

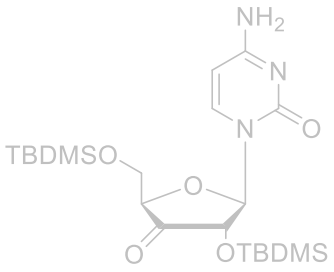
2',5'-Bis-O-(*tert*-butyldimethylsilyl)-3'-oxocytidine)



9.

2',5'-Bis-O-(*tert*-butyldimethylsilyl) cytidine **(8)** (200mg, 0.423mmol) was suspended in MeCN: H_2O (1:1, 7 mL). This mixture was kept under argon and stirred. TEMPO (2.5 mg, 0.016 mmol) and BAIB (423 mg,0.89 mmol) were added to the reaction mixture stirred for 2 hours under argon. TLC analysis (EtOAc:MeOH, 8:2) showed presence of starting material at (R_f = 0.7) and another faint spot at (R_f = 0.2). The mixture was extracted using DCM (4 x 10mL). The resulting organic layers were combined and washed with H_2O. TLC analysis indicated starting material (R_f = 0.7) and product spot (R_f = 0.2) in both aqueous and organic layers. The organic layer was dried over $MgSO_4$ and filtered. The aqueous layers were combined and cooled over ice. Dropwise acidification with 2 M aqueous HCl was carried out until the solution reached a pH of 2. The resulting precipitate was filtered, and the residue was washed using H_2O (10mL). This, however, did not yield the desired product.

2’,5’-Bis-O-(*tert*-butyldimethylsilyl)-3’-oxocytidine) (general procedure)



9.

2’,5’-Bis-O-(*tert*-butyldimethylsilyl) cytidine **(8)** (300mg, 0.635 mmol) was dissolved in anhydrous DCM (2 mL) under argon. The reaction mixture was cooled to 0 °C and a suspension of Dess-Martin periodinane (429mg, 1.01 mmol) in anhydrous DCM (6.9 mL) was added dropwise whilst stirring. This mixture was stirred under argon at 0°C for 15 minutes. After this time, it was allowed to gradually warm to room temperature and was stirred for 3 hours. TLC analysis (DCM: MeOH, 90:10) showed presence of starting material at (R_f=0.3) and formation of a possible product at (R_f=0.6). The reaction mixture was diluted using ethyl acetate (10mL). This mixture was washed with $NaHCO_3$ (10mL), H_2O (10mL) and brine (12mL). The organic layer was dried over $MgSO_4$ and the solvent was removed under reduced pressure. This reaction afforded recovered starting material.

5. References

1. Bates, Sarah A. “Deoxyribonucleic Acid (DNA).” Genome.gov, November 23, 2023. https://www.genome.gov/genetics-glossary/Deoxyribonucleic-Acid
2. Bates, Sarah A “Phosphate Backbone.” Genome.gov, November 22, 2023. https://www.genome.gov/genetics-glossary/Phosphate-Backbone
3. Hakem R. DNA-damage repair; the good, the bad, and the ugly. EMBO J. (4):589-605. (2008)
4. **Figure 1:** Image created using Biorender.com
5. Deans AJ, West SC. DNA interstrand crosslink repair and cancer. Nat Rev Cancer.; **11**, 467–480. (2011)
6. Enoiu M, Jiricny J, Schärer OD. Repair of cisplatin-induced DNA interstrand crosslinks by a replication-independent pathway involving transcription-coupled repair and translesion synthesis. Nucleic Acids Res. **40**, 8953-64, (2012)
7. Stone MP, Cho YJ, Huang H, Kim HY, Kozekov ID, Kozekova A, Wang H, Minko IG, Lloyd RS, Harris TM, Rizzo CJ. Interstrand DNA cross-links induced by alpha,beta-unsaturated aldehydes derived from lipid peroxidation and environmental sources. Acc Chem Res.; **41(7)**: 793-804, Jul (2008)
8. Kozekov ID, et al. DNA interchain cross-links formed by acrolein and crotonaldehyde. *J. Am. Chem. Soc.* **125** :50–61, (2003)
9. Lopez-Martinez D, Liang CC, Cohn MA. Cellular response to DNA interstrand crosslinks: the Fanconi anemia pathway. Cell Mol Life Sci, **73(16),** 3097-114, Aug (2016)
10. **Figure 2:** Image created using Biorender.com
11. Rageul J, Kim H. Fanconi anemia and the underlying causes of genomic instability. Environ Mol Mutagen.; **61(7)**, 693-708, (2020)
12. Chválová K, Brabec V, Kaspárková J. Mechanism of the formation of DNA-protein cross-links by antitumor cisplatin. Nucleic Acids Res.; **35(6)**: 1812-21, (2007)
13. Shen DW, Pouliot LM, Hall MD, Gottesman MM. Cisplatin resistance: a cellular self-defense mechanism resulting from multiple epigenetic and genetic changes. Pharmacol Rev. **64(3):**706-21, (2012)
14. Fang Can-Bin, Wu Hua-Tao, Zhang Man-Li, Liu Jing, Zhang Guo-Jun, Fanconi Anaemia Pathway: Mechanisms of Breast Cancer Predisposition Development and Potential Therapeutic Targets, Frontiers in Cell and Developmental Biology, **2(8)**: 160, (2020)

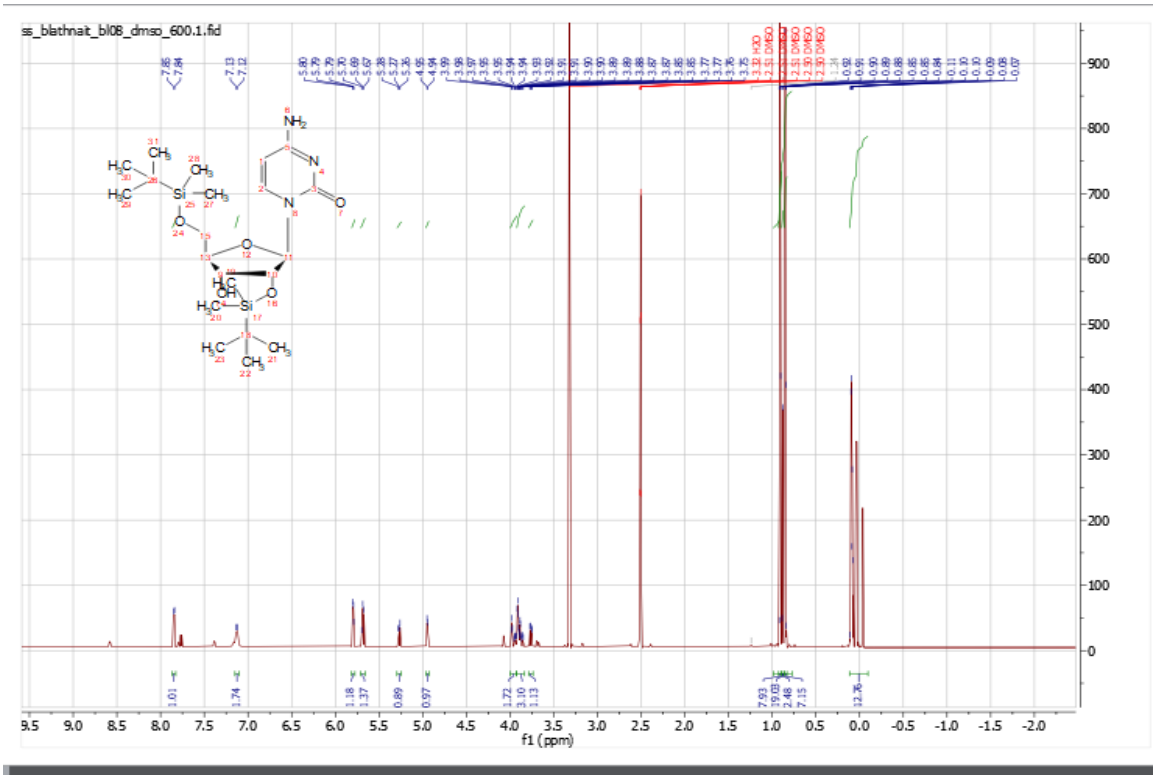
15. Mason JM, Sekiguchi JM. Snm1B/Apollo functions in the Fanconi anemia pathway in response to DNA interstrand crosslinks. Hum Mol Genet. **20(13)**:2549-59, (2011)
16. Duxin JP, Walter JC. What is the DNA repair defect underlying Fanconi anemia? Curr Opin Cell Biol.; **(37):**49-60, Dec (2015)
17. Allerston CK, Lee SY, Newman JA, Schofield CJ, McHugh PJ, Gileadi O. The structures of the SNM1A and SNM1B/Apollo nuclease domains reveal a potential basis for their distinct DNA processing activities. Nucleic Acids Res; **43(22)**: 11047-60, Dec (2015)
18. Buzon B, Grainger R, Huang S, Rzadki C, Junop MS. Structure-specific endonuclease activity of SNM1A enables processing of a DNA interstrand crosslink. Nucleic Acids Res.;**46(17**):9057-9066, (2018)
19. **Figure 3:** Allerston, Charles K, Crystal Structure of Human DNA Cross-Link Repair 1A, November 28, (2012). https://doi.org/10.2210/pdb4B87/pdb.
20. Sengerova. B, Allerston C.K., Abu M., Lee S.Y., Hartley J., Kiakos K., Schofield C.J., Hartley J.A., Gileadi O., McHugh P.J. Characterization of the human SNM1A and SNM1B/Apollo DNA repair exonucleases. J. Biol. Chem.; **287**:26254–26267, (2012)
21. Wang, A. T.; Sengerová, B.; Cattell, E.; Inagawa, T.; Hartley, J. M.; Kiakos, K.; Burgess-Brown, N. A.; Swift, L. P.; Enzlin, J. H.; Schofield, C. J., Human SNM1A and XPF–ERCC1 collaborate to initiate DNA interstrand cross-link repair. Genes Dev, **25 (17)**, 1859-1870, (2011)
22. Wang LC, Gautier J. The Fanconi anemia pathway and ICL repair: implications for cancer therapy. Crit Rev Biochem Mol Biol, **45(5)**:424-39, Oct (2020)
23. Jordheim, L. P., Barakat, K. H., Heinrich-Balard, L., Matera, E.-L., Cros-Perrial, E., Bouledrak, K, Dumontet, C, Small Molecule Inhibitors of ERCC1-XPF Protein-Protein Interaction Synergize Alkylating Agents in Cancer Cells. Molecular Pharmacology, **84(1)**, 12–24. (2013)
24. Lawley PD, Phillips DH. DNA adducts from chemotherapeutic agents. *Mutat Res* **355**:13–40, (1996)
25. Magaña-Schwencke N, Henriques JA, Chanet R, et al. The fate of 8-methoxypsoralen photoinduced crosslinks in nuclear and mitochondrial yeast DNA: comparison of wild-type and repair-deficient strains. *Proc Natl Acad Sci U S* A: **79:** 1722–1726, (1982)
26. Fay E.M, McGouran J.F, Exploring the binding pockets of the DNA damage repair enzyme SNM1A through nucleobase modification, Results in Chemistry, **Vol 5**, 100939, (2023)

27. Lee S.Y. Cephalosporins inhibit human metallo beta-lactamase fold DNA repair nucleases SNM1A and SNM1B/apollo. *Chem. Commun. (Camb.)***52**:6727–6730, (2016)
28. Wu, S. H., Hsiao, Y. T., Chen, J. C., Lin, J. H., Hsu, S. C., Hsia, T. C., Yang, S. T., Hsu, W. H. & Chung, J. G. Bufalin alters gene expressions associated DNA damage, cell cycle, and apoptosis in human lung cancer NCI-H460 cells in vitro. Molecules **19**, 6047–6057 (2014)
29. Dürr EM, McGouran JF. Probing the Binding Requirements of Modified Nucleosides with the DNA Nuclease SNM1A. Molecules**, 26(2):320** Jan (2021)
30. Shekhar, C. 'A Guide to Activity Based Probes', The Scientist, www.thescientist.com, Sept (2006)
31. Fonović M, Bogyo M. Activity-based probes as a tool for functional proteomic analysis of proteases. Expert Rev Proteomics**, 5(5)**:721-30, Oct (2008)
32. Isobe H., Fujino T. Triazole-linked analogues of DNA and RNA (^{TL}DNA and ^{TL}RNA): Synthesis and functions. *Chem. Rec.* **14(1)**:41-51 *(*2014)
33. Berney M, Doherty W, Jauslin WT, T Manoj M, Dürr EM, McGouran JF. Synthesis and evaluation of squaramide and thiosquaramide inhibitors of the DNA repair enzyme SNM1A. Bioorg Med Chem. **46**:116369, Sept (2021)
34. Fay EM, Newton A, Berney M, El-Sagheer AH, Brown T, McGouran JF. Two-Step Validation Approach for Tools to Study the DNA Repair Enzyme SNM1A. Chembiochem., 24**(13)**, Jul (2023)
35. E. M. Tacconi, S. Badie, G. De Gregoriis, T. Reisländer, X. Lai, M. Porru, C. Folio, J. Moore, A. Kopp, J. B. Torres, *EMBO Mol. Med.*, **11**, e9982, (2019)
36. Bartoszewicz A., Kalek M., Nilsson J., Hiresova R., Stawinski J, *Synlett*, 37-40. (2008)
37. J. Hejna, S. Philip, J. Ott, C. Faulkner, R. Moses, *Nucleic Acids Res.* (2007)
38. Abdullah, U. B.; McGouran, J. F.; Brolih, S.; Ptchelkine, D.; El-Sagheer, A.H.; Brown, T.; McHugh, P. J., RPA activates the XPF-ERCC1 endonuclease to initiate processing of DNA interstrand crosslinks. The EMBO journal, **36 (14)**, 2047-2060, (2017)
39. Eckstein F. Phosphorothioates, essential components of therapeutic oligonucleotides. Nucleic Acid Ther.;**24(6)**:374-87, Dec (2014)
40. Kelvin K, Ogilvie K, The tert-Butyldimethylsilyl Group as a Protecting Group in Deoxynucleosides. Canadian Journal of Chemistry. **51(22)**: 3799-3807, (1973)
41. **Fig 5** Image created using Biorender.com
42. **Fig 7** Image created using biorender.com

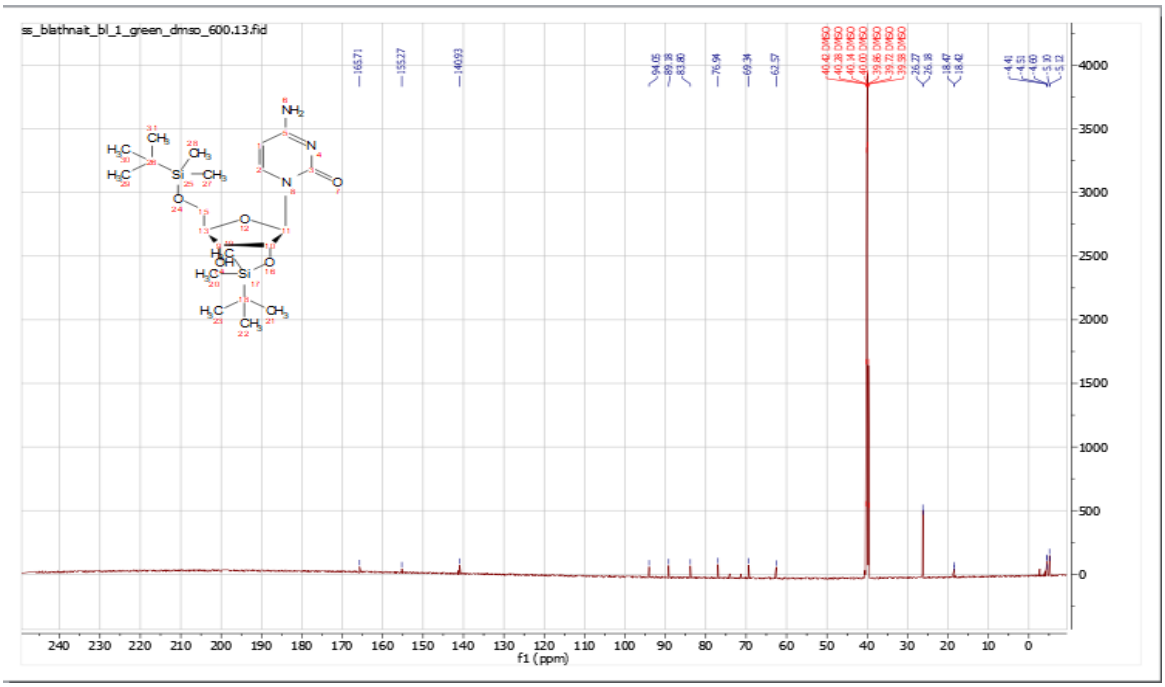
43. Ogilvie, K. K., Schifman, A. L. & Penney, C. L. The synthesis of oligoribonucleotides. III. The use of silyl protecting groups in nucleoside and nucleotide chemistry. VIII. Can. J. Chem. **57**, 2230–2238 (1979).
44. Reddy, M.; Hanna, N. B.; Farooqui, F., Ultrafast cleavage and deprotection of oligonucleotides synthesis and use of CAc derivatives. Nucleosides Nucleotides Nucleic Acids, **16 (7-9)**,1589-1598. 1997
45. Takeda, Y., Ohlendorf, D. H., Anderson, W. F. & Matthews, B. W. DNA-binding proteins. Science. **221**, 1020–1026 (1983).
46. McCall KA, Huang C, Fierke CA. Function and mechanism of zinc metalloenzymes. J Nutr.;**130(5S Suppl):**1437-46, May (2000)
47. Robert J.D, Caserio M.C, *Basic Principles of Organic Chemistry,* **(2)** (1977)
48. Sengerova, B.; Wang, A. T.; McHugh, P. J., Orchestrating the nucleases involved in DNA interstrand cross-link (ICL) repair. Cell Cycle, 10 **(23)**, 3999-4008. (2011)
49. Wasner, M., Arion, D., Borkow, G., Noronha, A., Uddin, A. H., Parniak, M. A.& Damha, M. J. Physicochemical and biochemical properties of 2’,5’-linked RNA and 2’,5’-RNA:3’,5’-RNA “hybrid” duplexes. Biochemistry **37**, 7478–7486 (1998).
50. Prakash, C., Saleh, S. & Blair, I. A. Selective removal of phenolic and alcoholic silyl ethers. Tetrahedron Lett. **35**, 7565–7568 (1994).
51. 18. Sawai, H., Ishibashi, K., Itoh, M. & Watanabe, S. 2′,5′-Oligoadenylate and 2′,5′-oligoadenylate phosphodiesterase in human plasma. Biochem. Biophys. Res.Commun. **125**, 1061–1066 (1984).
52. Chandrasekaran AR, Mathivanan J, Ebrahimi P, Vilcapoma J, Chen AA, Halvorsen K, Sheng J, Hybrid DNA/RNA nanostructures with 2'-5' linkages. Nanoscale. 5;**12(42)**:21583-21590 (2020)

6. Appendix

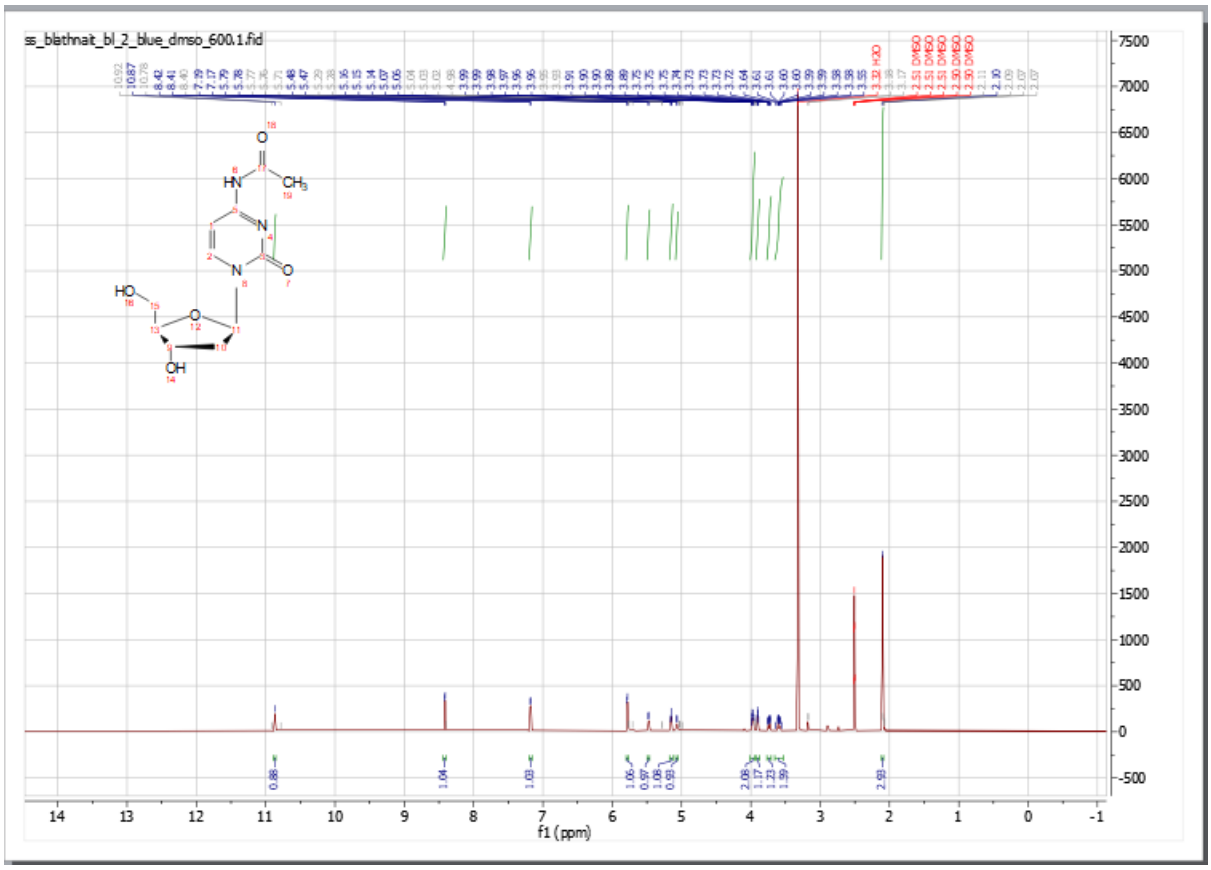
1.1 1H NMR for **8**



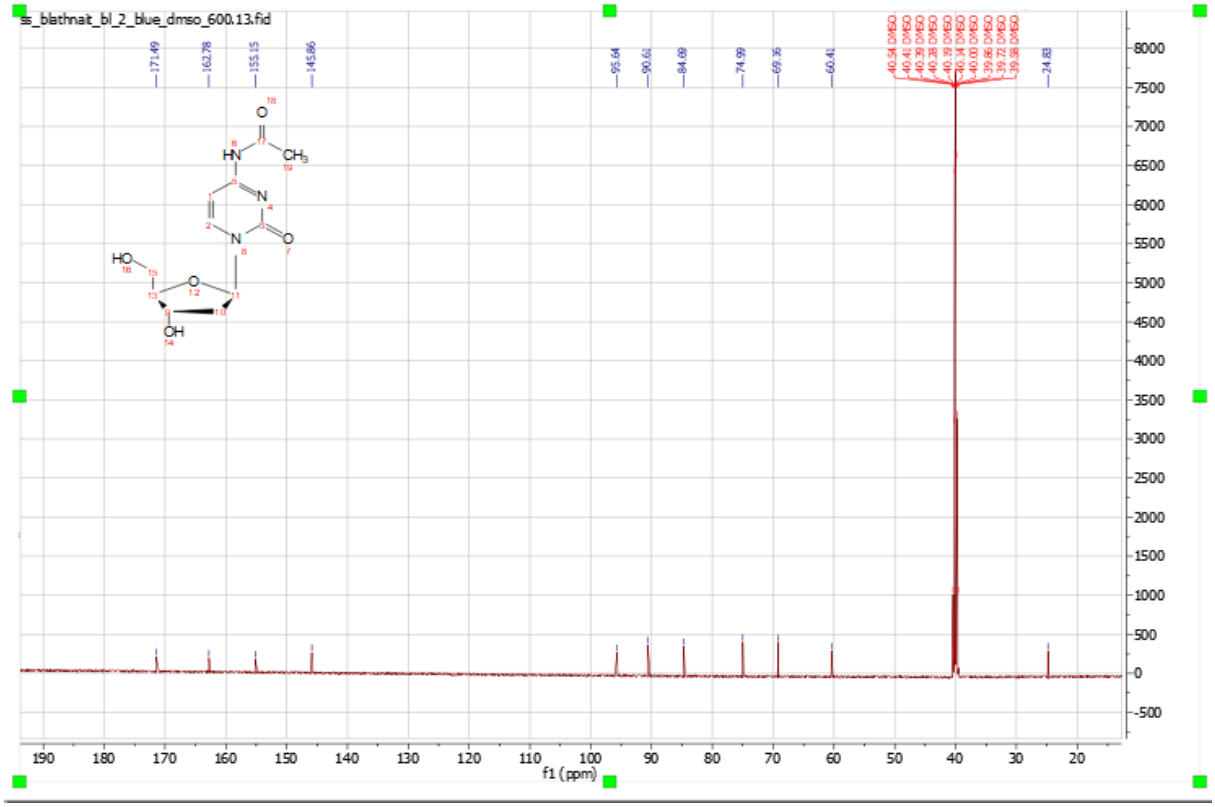
1.2 ^{13}C NMR for **8**



2.1 ^{1}H NMR for 10



2.2 ^{13}C NMR for 10



3.1 13C NMR of N4-Acetyl-2',5'-Bis-O-(*tert*-butyldimethylsilyl) cytidine

